

Isolated Level-shifting in Monolithic GaN

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Abstract—This paper presents the concept development - IC design and simulation - of two galvanically isolated, digital level shifter topologies in monolithic GaN. The level shifter is aimed to be used as a part of a gate driver for a half-bridge. Both topologies are compared in terms of attainable Common-Mode Transient Immunity (CMTI), propagation delay, current consumption, and estimated area. The two topologies use different methods to handle CMTI, the details can be found in [1] and [2]. The IC design of the topologies in this paper differs slightly due to a lack of d-mode HEMT's in the PDK used. Both level shifters need to translate the voltage from the low-side domain (0 - 6 V) to the high-side domain (80 - 86 V), while achieving sufficient CMTI - $|dV_{SSH}/dt| > 100$ V/ns - and sufficiently low propagation delay - $t_d < 10$ ns. The circuits are implemented in imec's gan200 technology, and their operation was simulated up to 2 MHz.

I. INTRODUCTION

In order to simultaneously support the increase in society's electrification and the decrease in carbon emissions, more efficient power conversion technologies are needed. For this purpose, wide-bandgap semiconducting materials such as Silicon Carbide (SiC) and Gallium Nitride (GaN) have been introduced to produce power transistors.

In addition, GaN has lower C_{ISS} when compared to Si, which means it can switch faster. High switching speeds lead to fast common-mode transients, which pushes the need for extremely robust level shifting techniques to safely translate signals from low-voltage levels to high-voltage levels. Moreover, the application in question requires galvanic isolation via e.g. capacitors or transformers - but following the trend of monolithic GaN integration, these requirements pose quite a challenge. In this paper, the described challenge is tackled.

A. Why GaN?

As the level shifter must operate in the voltage range of 0 - 6 V, the devices used in the level shifter topologies must have high enough breakdown voltage. The wurtzite structure of GaN crystal lattice has stronger chemical bonds, leading to the higher critical electric field, and thus the higher breakdown voltage of GaN-based device compared to the breakdown voltage of an Si-based device.

Furthermore, regarding the performance comparison between GaN- and Si-based devices, two important figures of merit (FOM's) must be taken into account:

- 1) $R_{DS,on} \cdot Q_{gate}$ - for the same on-resistance, GaN exhibits lower gate charge than Si (shown in Figure 1), making it suitable for faster switching (but this may be a drawback when the IC has a lot of ringing);
- 2) $R_{DS,on} \cdot Area$ - for the same on-resistance, GaN-based design can be more compact than Si-based design.

B. Solution Specifications

Level shifter specifications to be met are given in Table I. As may be seen, the main design constraints include sufficient speed (i.e. the delay must be lower than the values stated in

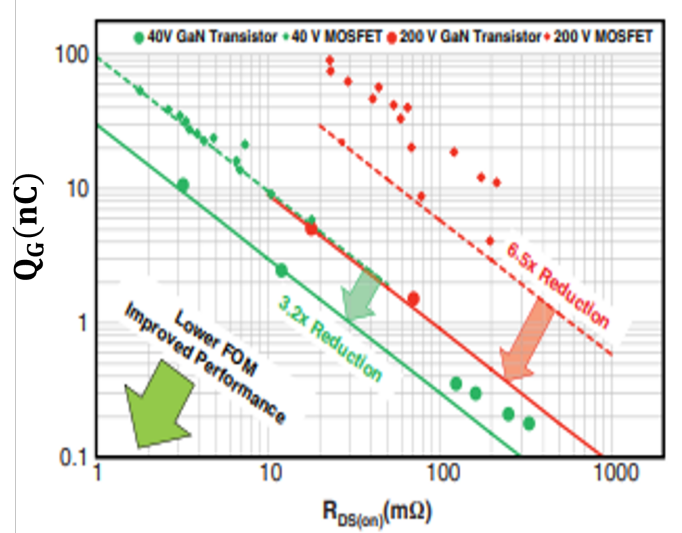


Fig. 1: "On-resistance vs. total gate charge comparison for Si- and GaN-based 40 V and 200 V power devices." [3]

Specification	Value	Unit
Technology	imec gan 200 V	-
$V_{in,max}^*$	80	V
f_{sw}	< 2	MHz
CMTI	100	V/ns
Propagation delay (pre-PEX)	< 10	ns
Propagation delay (post-PEX)	< 15	ns
Propagation delay matching	< 3	ns
Minimum on-off times	25	ns
Maximum on-off times	500	ns

Table I: Level Shifter Specifications.

the referenced Table) and the ability to function even when unexpected, rapid voltage signal transients occur (characterized by CMTI - Common Mode Transient Immunity). After satisfying these initial constraints, optimizations regarding power consumption and area overhead can be executed.

C. Workflow

As a first step towards reaching the defined goal, a literature study is needed. The designer should understand not only the motivation behind the use of GaN as a semiconducting material but also the operating principle of GaN-based transistors. After grasping the fundamentals, a study on circuit topology of the level-shifter may be done - which topology would best suffice and which potential changes to it are needed.

When the appropriate topology is selected, the actual concept development may begin; circuit design and layout may be executed. After the circuit layout has been implemented, post-layout simulations are to be executed to further verify the functionality of the proposed design. After this step, the circuit is ready for fabrication and measurements.

In the following paper, solution steps up to (and excluding) layout design are elaborated on.

II. LITERATURE STUDY

A. GaN Transistors [3]

Gallium nitride exhibits an interesting hexagonal, crystal structure called wurtzite [3]. If the lattice is subjected to strain, its deformation will lead to an electric field generation proportional to the strain applied. To achieve this, a thin layer of AlGaN on top of the GaN crystal may be grown [3]. At the interface of these two layers, a so called two-dimensional electron gas (in short: 2DEG) will be induced [3]. Hence, a reservoir of highly conductive, confined electrons is formed, providing a basis for the high electron mobility transistor (HEMT), the primary component used for the design described in this work.

As with any Field Effect Transistor (FET), HEMT's are comprised of four electrodes: bulk, drain, gate and source electrode. The source and drain electrodes pierce through AlGaN to make an ohmic contact with the 2DEG and allow for a current flow through the transistor [3]. In order to control this current flow, a gate electrode is placed on top of the AlGaN layer [3].

There are two main types of HEMT's – a depletion-mode and an enhancement-mode transistor. The difference is as follows (explanation based on [3]): the depletion-mode HEMT conducts current when a zero bias is applied to its gate; in other words, it turns off only when a negative bias gate voltage is applied. On the other hand, the enhancement-mode HEMT behaves in such a way that it is off when the bias gate voltage is zero, and it conducts current when a positive bias gate voltage is applied. The technology node used in this project is restricted to the enhancement-mode HEMT's; therefore, only this type of transistor will be referred to in further text and figures.

B. Level Shifters

A level shifter is an electronic circuit used to translate a signal from one voltage level to another, such that devices with different supply voltages may communicate safely. In essence, it is used to bridge different voltage domains, and ensure that both low-side and high-side components remain functional as well as robust to any damage.

Level shifters are commonly used between a micro-controller and sensor interfaces, in communication busses (I^2C , SPI, UART), in memory cards and embedded devices, but also in power electronics' gate drivers - the application this work focuses on.

One may categorize level shifters in various ways, e. g.:

- Direction-based classification: if the signal flows only one way (from input to output), a level shifter is unidirectional; on the contrary, if the signal flows in both directions (connected devices transmit and receive), a level shifter is bidirectional. This work focuses on a unidirectional level shifter design.
- Implementation-based classification: simple level shifters might be designed on the basis of resistive division, while more complex level shifters are dedicated integrated circuits comprised of a low-side part and a high-side part with a capacitive/transformer isolation. The topology used in this work will be elaborated in sections III and IV.

III. CIRCUIT TOPOLOGY 1 AND SIMULATION RESULTS

The first proposed level shifter topology based on [1] is shown in Figure 2 and is discussed in this section.

As stated in [1] and seen in Figure 2, the topology in question utilizes two techniques which work hand-in-hand to prevent the overcharge of the high-side power supply capacitor during high (negative) dV/dt transitions and improve CMTI. **Note:** in the following paragraphs of this section, all the signals, nodes, and components mentioned are the ones seen in Figure 2, if not stated otherwise.

The first technique employed is the so-called Switching Node Falling Edge Detection (SNFED). In short, the operating principle is as follows: during the falling edge of V_{SSH} , the common mode displacement current I_{CMDC} is generated and flows through C_B , generating an overshoot at N_3 (N_4). The sense blocks (inverters) detect this change, and the SNFED-block generates a blanking signal V_B to suppress the faulty logic signal by turning on the transistor E_7 (E_8), such that I_{CMDC} flows safely to V_{SSH} . [1] may be consulted for more details.

The main bottleneck of SNFED is the turning speed of E_7 (E_8), which is limited by the propagation delay of the digital loop [1]. This means that SNFED does not entirely help in the case of high dV/dt transitions (meaning $50V/ns$ and above), such as in half-bridge GaN drivers where the level shifter from this work is to be used. That is why a second optimization is needed, namely the Dynamic Discharge Control (DDC) circuit.

DDC operates on the following principle: when a high negative dV_{SSH}/dt occurs, I_{CMDC} is generated and injected to the high-side part via capacitors C_{SENSE} . This current will normally flow through diodes $E_{17,19}$ ($E_{18,20}$) and transistor E_{13} (E_{14}), and the voltage V_A (V_{nodeB}) will rise such that transistor E_{11} (E_{12}) turns on and enables the discharge of C_B to V_{SSH} . In the case of positive dV_{SSH}/dt , C_{SENSE} is charged by the current path through the transistor E_{15} (E_{16}). [1] may be consulted for more details.

A few notable changes with respect to the topology described in [1] are:

- diodes $E_3 - E_6$, which replace transistors $M_{5,6}$ from Figure 6 in [1] - in this way, nodes N_3 and N_4 are not pulled low constantly, but only after the voltage peaks needed for sensing inverters occur (otherwise the circuit malfunctions);
- resistors R_1 and R_2 are added to ensure a high enough voltage value and a long enough RC-discharge constant τ , such that sensing inverters are able to capture any change at nodes N_3 and N_4 ; resistors $R_3 - R_6$ are added after the execution of OCC (Operating Condition Check);
- transistors M_9 and M_{10} from Figure 6 in [1] are omitted since they do not contribute to this topology;
- diodes in the low-side part are added to sink I_{CMDC} during high positive dV/dt and ensure that the voltages over low-side transistors do not cross the breakdown threshold.

A. Logic Gates

As is the case with the majority, if not all of the level shifter topologies, topologies used in this work incorporate logic gates such as inverters, NAND-gates, and NOR-gates. Hence, the first step towards a fully functional level shifter topology is designing sufficient logic gates.

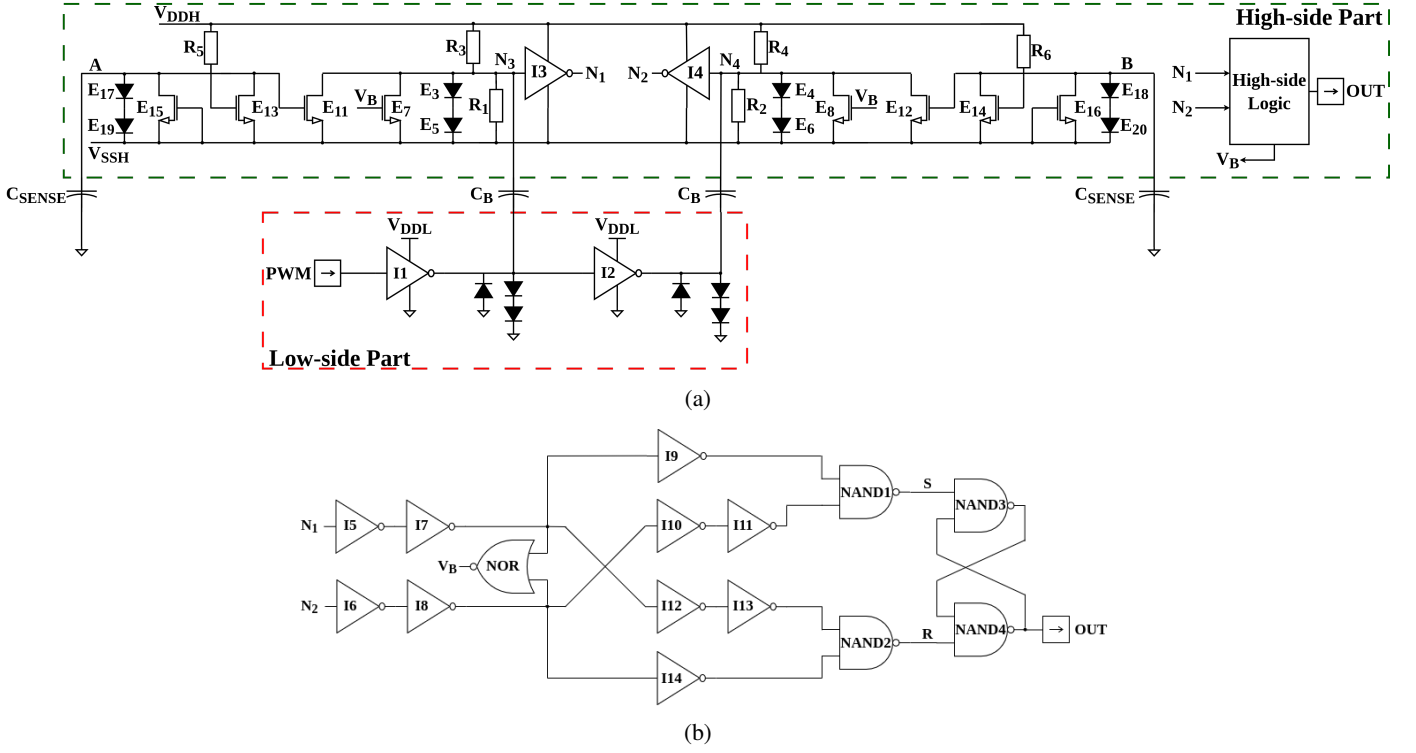


Fig. 2: (a) Capacitive level shifter topology based on [1]. (b) High-side logic comprised of the Fault Logic Blanking (FLB) and a Set-Reset Flip-Flop (SRFF), respectively.

Taking into account the limitation of the technology used in this work, that is, not having a p-type e-mode HEMT, one might intuitively opt for the simplest possible inverter network, where the pull-down network consists of a single n-type HEMT, and the pull-up network is a resistor. This topology is shown in Figure 3a and from now on, it will be referred to as the *Type 1* inverter. Area-wise, this design is extremely compelling if the resistance needed is low enough, but it might not be as feasible regarding its operation speed and power consumption.

Therefore, an optimized topology is considered, as depicted in 3b and based on [4], but slightly modified. This *Type 2* inverter works the following way: the pull-down network naturally switches on when the input voltage V_{in} is high, but in the meantime the diode-connected E_1 charges the capacitor C_b . When V_{in} switches from high to low, then $V_{G2} = 2V_{DD} - V_{TH1}$ holds due to the previously charged capacitor. From this, it follows $V_{GS2} = 2V_{DD} - V_{TH1} - V_{OUT}$, which ensures that E_2 can pull V_{out} high when V_{in} is low.

The results needed for comparison are given in Table II. As may be observed, the propagation delay and current consumption are somewhat lower for the *Type 2* inverter than for the *Type 1* inverter. Moreover, the estimated area is lower for the *Type 2*, which is due to overhead introduced to the resistor in the *Type 1* - the minimal value allowed by the technology used is too high, so multiple parallel segments must be used. For these reasons, at this stage the *Type 2* inverter is a preferred topology, hence other logic gates will be derived from it.

B. Simulation Results

Now that the basic building blocks have been implemented, the remainder of the level shifter circuit was designed such that all requirements are satisfied. This was done by iteratively sizing the circuit and simulating its behavior until the expected operation was achieved.

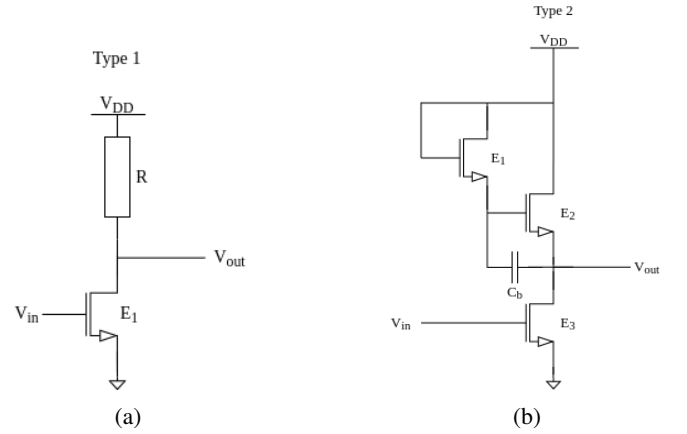


Fig. 3: Inverter topologies to be compared, where the pull-up network is implemented by means of a (a) resistor, (b) charge-pump circuit [4].

Specification	Value (<i>Type 1</i>)	Value (<i>Type 2</i>)	Unit
t_{rise}	1.35	0.92	ns
t_{fall}	1.03	1.09	ns
t_{delay}	1.19	1.01	ns
I_{DC}	363.54	316.11	μA
A_{est}	5730	2310	μm^2

Table II: Simulation results for inverter topologies.

Due to the nature of the level-shifting circuit, the primary focus is on the transient analysis results. Nevertheless, a DC simulation was executed to verify whether each node has a proper starting point (mainly a logical one or a logical zero). After confirming the DC operating point was valid, transient analysis is performed. As the circuit operating frequency is 2MHz, a conservative transient analysis lasts for 1.1 μs to capture two periods of the output signal. The analysis was done

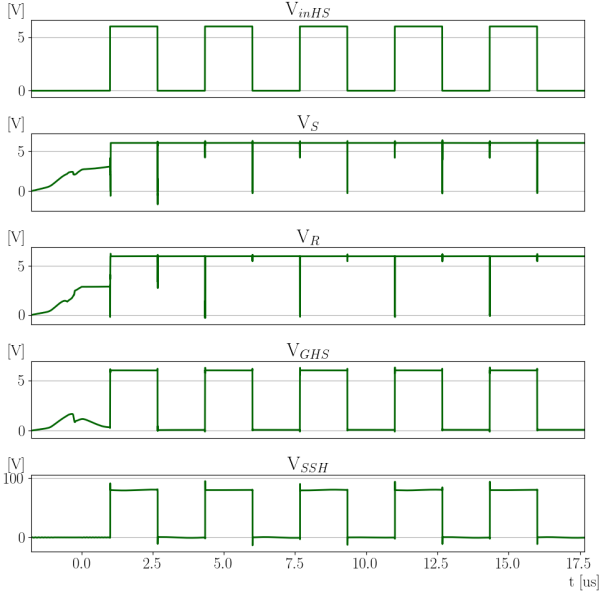


Fig. 4: Transient analysis results for the half-bridge configuration. V_{INHS} and V_{SSH} are referenced to (low-side) ground (0V), while V_S , V_R and V_{GHS} are referenced to V_{SSH} .

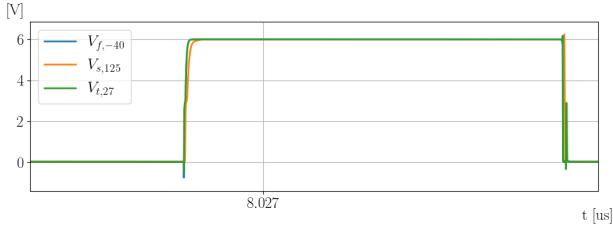


Fig. 5: Corner analysis results for the level shifter output in the half-bridge configuration, for the nominal case (V_{nom}) and worst-case corners: s , 125°C (V_{s125}), and f , -40°C (V_{f-40}).

for a maximum absolute dV_{SSH}/dt of 100V/ns to ensure that the circuit satisfies given specifications.

The transient analysis and the corner analysis were run on an isolated level shifter with a simple capacitive load, but in a half-bridge configuration as well. This configuration mimics a real circuit environment with CMTI up to 100V/ns , where the level shifter should be able to work properly, and is shown in Figure 6 - cap_lvsh_LS corresponds to the **Low-side Part** and capacitors in Figure 2, while cap_lvsh_HS corresponds to the **High-side Part** in Figure 2.

The results of the transient analysis are given in Figure 4, for a larger number of cycles to ensure the circuit is functional. It may be observed that the level shifter does indeed operate as intended - the output signal follows the input signal with a slight delay, and in the event of high dV_{SSH}/dt , the output signal is not corrupted.

In this configuration, corner analysis was done using standard PDK models, but also using Operating Condition Check (OCC) models. OCC is done to confirm that the circuit operates within design specifications, respecting all safety guidelines (e. g. V_{DS} should not exceed the maximum nor fall below the minimum allowed value for any transistor in the circuit).

The results of the corner analysis (for the voltage V_{GHS}) for the worst-case corners (f , -40°C , s , 125°C) are shown in Figure 5.

The final check to do before proceeding to the layout design is the power density check. For each transistor, one must manually check whether the power density of said transistor is low enough - in this work, it should be lower than (or equal to) 1W/mm^2 . The power density is calculated using Equation 1, where $\overline{V_{DS}}$ is the average drain-source voltage of the transistor, $\overline{I_D}$ is the average drain current of the transistor, and A_{est} is the estimated area of the transistor when instantiated in *Layout XL*.

$$\text{Power density} = \frac{\overline{V_{DS}} \overline{I_D}}{A_{est}} \quad (1)$$

An important conclusion was drawn during this check - in many instances, *Type 2* inverter presented earlier exhibits a power density that is an order of magnitude (or more) higher than the maximum allowed power density. Even if the sizes of transistors were enlarged, this would result in additional current consumption, hence the power density issue would hardly be solved while maintaining a reasonable total current consumption.

IV. CIRCUIT TOPOLOGY 2 AND SIMULATION RESULTS

The second proposed level shifter topology based on [2] is shown in Figure 7 and is discussed in this section.

As stated in [2] and seen in Figure 7, the topology in question utilizes a few techniques to ensure sufficient CMTI and improve mismatch tolerance of the level shifter. **Note:** in the following paragraphs of this section, all the signals, nodes, and components mentioned are the ones seen in Figure 7, if not stated otherwise.

The first technique in question - FeedBack Discharge Control (FBDC) - is almost identical to the DDC method described in Section III and based on [1], hence its operation principle will be omitted here. For more details, see [2].

Furthermore, Common-Mode Blanking Pulse (CMBP) is employed to mitigate the false trigger of the high-side logic in the event of a negative common-mode dV_{SSH}/dt . A pulse V_B is generated (via instance *I16*) to pull V_{N3} (V_{N4}) low by turning on transistor E_9 (E_{10}), and a pulse V_{CM} is generated (via instances *I17*, *I18*, *NOR*) to prevent any residual, faulty logic states from reaching S/R nodes, leaving the output signal unchanged. For more details, see [2].

Finally, MisMatch Reduction Circuit (MMRC) reduces mismatch effects during positive dV_{SSH}/dt . In a real circuit, symmetrical components (e. g., transistors E_5 and E_6 in Figure 7a) which are ideally equal, slightly differ. Therefore, positive common-mode pulses at nodes N_3 and N_4 might not occur simultaneously. The circuit may wrongly interpret this difference in the common-mode signals as a differential signal, and the false triggering of the SRFF may occur. Hence, the two different common-mode signals should be equalized before reaching the SRFF to ensure functional circuit operation. This is done by generating a positive voltage pulse at the gate of E_{MMRC} (via instances *I21*, *I22*, *NA7*, *I23*), thus equating voltages at nodes N_1 and N_2 . For more details, see [2].

A few notable changes with respect to the initial topology described in [2] were made and are listed as follows:

- transistors M_3 , M_5 , M_{10} and M_{12} from [2] are omitted: M_3 , M_{10} due to lack of p -type devices, and M_5 , M_{12} because they would be constantly on, pulling nodes N_3 and N_4 low when not needed;

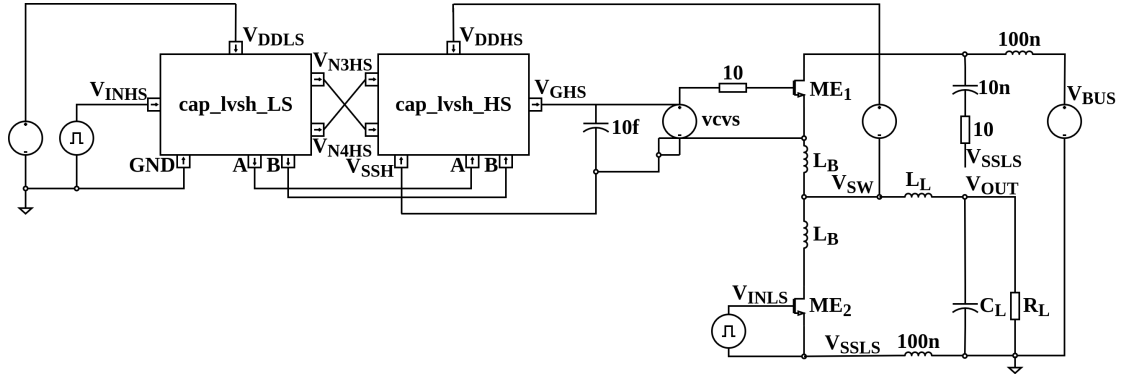


Fig. 6: Schematic of the half-bridge configuration for simulating level shifters in a real circuit environment, with CMTI up to 100 V/ns .

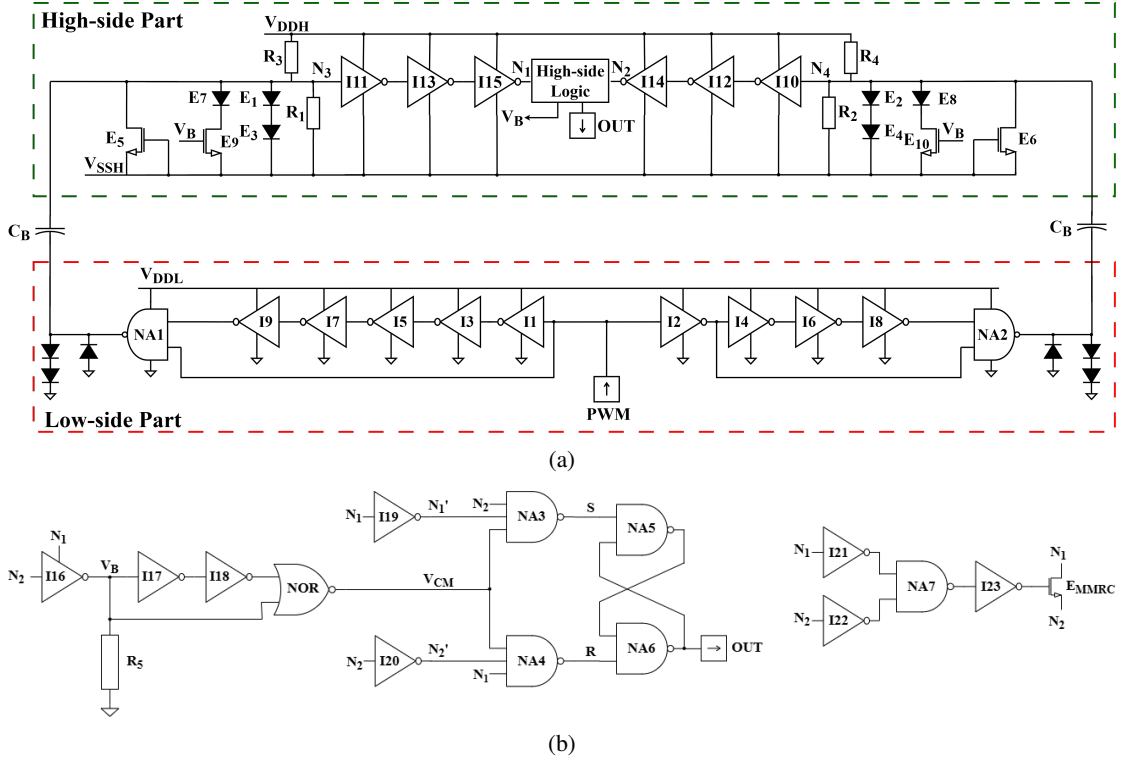


Fig. 7: (a) Capacitive level shifter topology based on [2]. (b) High-side logic comprised of the Common-Mode Blanking Pulse (CMBP), a Set-Reset Flip-Flop (SRFF), and the MisMatch Reduction Circuit (MMRC), respectively.

- diodes $E_1 - E_4$ and low-side diodes were added to provide sufficient current path in the event of large dV_{SSH}/dt transitions;
- resistors were added: R_1 and R_2 to ensure pulses V_{N3} and V_{N4} are large enough (exceed sensing inverters' thresholds) to be noticed by sensing inverters and forwarded to the rest of the logic, and R_3 and R_4 to ensure V_{N3} and V_{N4} do not fall out of the breakdown limits.

A. Simulation Results

Simulating the second level shifter topology was done by incorporating it into the same test-bench as shown in Figure 6. Therefore, only the final, most important results for the second topology will be presented.

The results of the transient analysis are given in Figure 8, for a larger number of cycles to ensure the circuit is functional. It may be observed that the level shifter operates as intended - the output signal follows the input signal with a slight delay, and in the event of a high dV_{SSH}/dt , the output signal is not affected.

There is one unexpected anomaly in the transient behavior of the set (S) node - during high positive dV_{SSH}/dt ($> 100 \text{ V/ns}$), it is pulled low, when it should not be. This shows that the design of the second level shifter topology should be further modified to address the issue mentioned.

The results of the corner analysis (for the voltage V_{GHS}) for the worst-case corners ($t, -40^\circ\text{C}$, and $t, 125^\circ\text{C}$) are shown in Figure 9. In addition, as was done for the first level shifter topology, some components of the second topology had to be resized to meet OCC and power density requirements. However, the topology was still not able to perform as expected across all corners, hence the design of this level shifter should be further reiterated and improved.

V. TOPOLOGY COMPARISON

The most important parameters - the highest simulated CMTI, the propagation delay, the total current consumption and the estimated area - were extracted from the final simulation ($t, 27^\circ\text{C}$ in the half-bridge configuration) for each topology, and

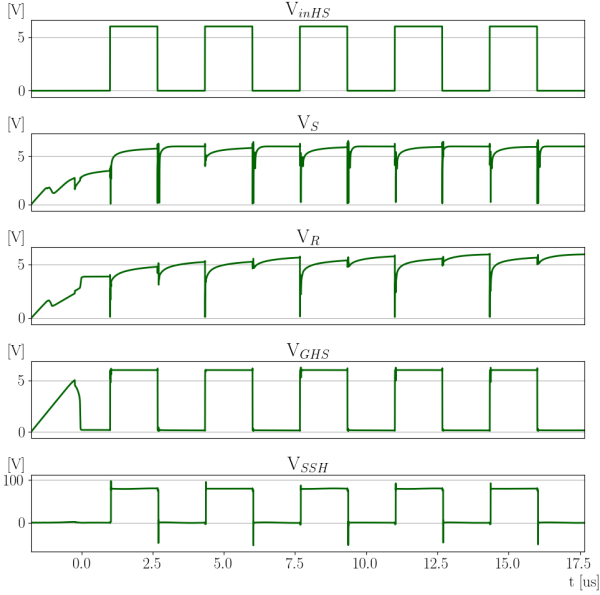


Fig. 8: Transient analysis results for the half-bridge configuration with the second level shifter topology. V_{INHS} and V_{SSH} are referenced to (low-side) ground (0V), while V_S , V_R and V_{GHS} are referenced to V_{SSH} .



Fig. 9: Corner analysis results for the level shifter output in the half-bridge configuration with the second level shifter topology, for the nominal case (V_{nom}) and worst-case corners: t , 125°C (V_{t125}), and t , -40°C (V_{t-40}).

are given in Table III, the pinnacle of this work. One may observe that both topologies successfully withstand voltage transients $|dV_{SSH}/dt| > 100 \text{ V/ns}$ - this is characterized by taking the minimum and maximum values of Equation 2.

$$CMTI = \frac{dV_{SSH}}{dt} \quad (2)$$

Moreover, it can be noted that the second level shifter topology has a significantly (nearly $\times 2$) smaller propagation delay in the case of output rising edge delay, as well as output falling edge delay. This might be counter-intuitive since the second topology utilizes more logic gates, so it is reasonable to assume its propagation delay would be longer than the propagation delay of the first topology. However, the first topology utilizes bigger logic gates and resistors (as resized during OCC and the power density check), which exhibit higher time constants, hence the total propagation delay is increased. This may also be surmised from the power consumption and estimated area - while the first topology consumes somewhat less current than the second topology (due to using fewer components than the second topology), its estimated area is quite close to the estimated area of the second topology (due to using bigger components than the second topology).

Specification	Value (Topology 1)	Value (Topology 2)	Unit
$CMTI_{min}$	-335	-273	V/ns
$CMTI_{max}$	324	300	V/ns
$t_{delay, rise}$	8.46	4.57	ns
$t_{delay, fall}$	9.65	5.57	ns
I_{DC}	8.62	9.43	mA
A_{est}	0.93	0.98	mm^2

Table III: Final results for both level shifter topologies extracted from the half-bridge configuration simulations (t , 27°C).

VI. CONCLUSION AND AKNOWLEDGEMENTS

The concept development - IC design and simulation - of two galvanically isolated, digital level-shifting circuits in monolithic GaN, used in the half-bridge GaN driver, was presented. As a pinnacle of the work, a comparison in terms of attainable Common-Mode Transient Immunity (CMTI), propagation delay, current consumption, and estimated area was documented (Table III). The first level shifter is based on the topology described in [1], which implements techniques such as DDC and SNFED to ensure a non-corrupted output signal in the event of high dV/dt transitions. The second level shifter is based on the topology described in [2], which utilizes circuits such as FBDC for sufficient CMTI, MMRC CMBP to block common-mode signals with improved mismatch tolerance. There were slight changes with respect to the topologies mentioned in [1] and [2] to successfully map the circuits to the GaN-technology used. Both level shifters translate the voltage from the low-side domain (0 - 6 V) to the high-side domain (80 - 86 V), while achieving sufficient CMTI - $|dV_{SSH}/dt| > 100 \text{ V/ns}$ - and sufficient propagation delay - $t_d < 10 \text{ ns}$. The level shifter topologies are implemented in imec gan200 technology, while their operation was simulated up to 2 MHz in CMTI-free environment and up to 300 kHz in the half-bridge configuration.

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